

ALPHY SAM RONI

Machine Learning Engineer · Data Scientist · AI Developer

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PROFESSIONAL SUMMARY

Motivated BCA student with hands-on internship experience in Machine Learning and AI at NextGenPro Innovation & Edupark Pvt. Ltd. Proven ability to build end-to-end AI systems — from computer vision pipelines and deep learning classifiers to NLP chatbots and multimodal assistants. Proficient in Python, TensorFlow, PyTorch, and Scikit-Learn. Google-certified AI Professional with expertise spanning AI Fundamentals, Brainstorming, Research, Communication, Content Creation, Data Analysis, and App Building. Passionate about applying data-driven solutions to real-world problems and eager to contribute to innovative engineering teams.

INTERNSHIP EXPERIENCE

Machine Learning Intern

April 2026 – June 2026

NextGenPro Innovation & Edupark Pvt. Ltd. · Ernakulam, Kerala

- Designed and trained classification models using Random Forest & SVM with real-world business datasets, achieving measurable accuracy improvements through feature engineering and hyperparameter tuning.
- Performed end-to-end data preprocessing & feature engineering — handling missing values, encoding, scaling, and dimensionality reduction — improving model performance across multiple datasets.
- Applied model evaluation techniques (cross-validation, confusion matrix, ROC-AUC) to validate and compare algorithm performance in production-ready workflows.
- Collaborated in an agile team environment with mentors on AI and Data Science research projects, strengthening understanding of ML lifecycle management.

EDUCATION

Bachelor of Computer Applications (BCA)

2024 – 2028

St. Thomas College, Thavalappara, Konni · Mahatma Gandhi University, Kottayam

Higher Secondary Education (CBSE)

Year of Passing: 2024

Holy Angels Model School, Pathanamthitta

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Google AI Professional Certificate — [Google / Coursera](#) | June 28, 2026

Completed 7-course program; built 20+ AI artifacts and a custom AI solution (vibe-coded). Proficient in effective prompting, output evaluation, and responsible AI usage for real workplace challenges.

- AI Fundamentals
- AI for Brainstorming and Planning
- AI for Research and Insights
- AI for Writing and Communicating
- AI for Content Creation
- AI for Data Analysis
- AI for App Building

AWS Certified Machine Learning Engineer – Associate (MLA-C01) — [Amazon Web Services](#) | June 27, 2026

Domain 2 Review completion certificate; covers ML engineering practices on AWS cloud infrastructure.

Artificial Intelligence Certification (6 Months) — [Institution](#) | 2024

PROJECTS

Catty — Multimodal AI Assistant

Technologies: [Streamlit](#) · [Gemini 2.5 Flash](#) · [YOLOv8](#) · [OpenCV](#) · [PyTorch](#) · [SQLite](#) · [SpeechRecognition](#)

- Architected a browser-based multimodal assistant supporting text, image, and voice interactions with a glassmorphism UI powered by Three.js animations.
- Integrated Google Gemini 2.5 Flash for conversational AI and Visual Question Answering via Salesforce BLIP, enabling rich cross-modal understanding.
- Implemented real-time object detection (YOLOv8) and image quality analysis (OpenCV); managed multi-session chat history with SQLite.

Vehicle Detection System

Technologies: [Python](#) · [YOLOv8](#) · [PyTorch](#) · [Google Colab](#)

- Built a real-time vehicle detection pipeline using YOLO; trained and evaluated on traffic datasets achieving accurate identification in images and video streams.
- Optimised inference speed for near-real-time processing, demonstrating strong computer vision and model deployment skills.

Animal Classification System

Technologies: [Python](#) · [TensorFlow](#) · [CNN](#) · [Deep Learning](#)

- Designed a CNN-based image classifier to categorise multiple animal species; performed data augmentation, preprocessing, training, and evaluation using TensorFlow.

Simple Medical Chatbot

Technologies: [Python](#) · [NLP](#) · [NLTK](#) · [Scikit-Learn](#) · [Pandas](#) · [NumPy](#)

- Developed an NLP-based chatbot for common healthcare queries, implementing text preprocessing, intent recognition, and automated conversational responses.

Face Detection System

Technologies: [Python](#) · [OpenCV](#) · [Haar Cascade](#) · [NumPy](#)

- Built a real-time face detection and tracking application using Haar Cascade classifiers and OpenCV image processing pipelines.

TECHNICAL SKILLS

Languages: [Python](#) · [Java](#) · [C](#) · [SQL](#)

ML / AI: [Supervised Learning](#) · [Deep Learning](#) · [Computer Vision](#) · [NLP](#) · [Random Forest](#) · [SVM](#) · [KNN](#) · [Decision Trees](#) · [Logistic Regression](#)

Frameworks & Libraries: [TensorFlow](#) · [PyTorch](#) · [Scikit-Learn](#) · [OpenCV](#) · [NLTK](#) · [Pandas](#) · [NumPy](#) · [Streamlit](#)

Tools & Platforms: [Google Colab](#) · [Git](#) · [GitHub](#) · [SQLite](#) · [YOLOv8](#) · [Three.js](#)

SOFT SKILLS

[Problem Solving](#) · [Team Collaboration](#) · [Communication](#) · [Leadership](#) · [Time Management](#) · [Critical Thinking](#) · [Adaptability](#)

LANGUAGES

[English](#) · [Malayalam](#) · [Hindi](#) · [Tamil \(Basic\)](#)